

Predicting 2025 NHL Free Agent Contracts

By: Jackson Chlebowy



Why Contract Prediction?



- Rapidly Rising Salary Cap
 - 2025: \$95.5m
 - 2026: \$104m
 - 2027: \$113.5m
- The NHL Contract Market is likely to experience a major reset, with players seeking more lucrative contracts
- Determining what past player stats and metrics contributed most to player contracts can help GMs make better decisions to utilize the rising cap best.

Project Objectives



- Create 2 separate models to predict 2025 NHL Free Agents contract length and AAV using regression and machine learning, based on traditional statistics, advanced metrics, and intangible factors
- Gain a better understanding of what factors most meaningfully contribute to an NHL player's contract length and Average Annual Value
 - Determine what stats or characteristics are most important for predicting a free agent's contract

Data Collection



Data Sources:

- PuckPedia – Contract Details
- Sportrac – Contract Details and FA List
- Hockey Ref – Traditional Season Statistics
- MoneyPuck – Advanced Metrics

Data Issues:

- Inconsistent Naming – For vars and players
- Inconsistent Data – multiple sources with conflicting or differently formatted data

Data Processing



Data Structure

- Collected 5 seasons' worth of past free agents (2019-2024) to train my models
- 3 Rows/Seasons worth of stats leading up to Free Agent Year
- 486 total players after filtering out players < 3 seasons in the NHL
- Calculated each player's Cap% at time of signing their previous contract

Collapsing The Data and standardizing statistics/60mins

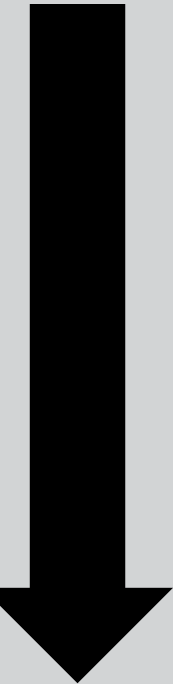
- Each player gets 1 row, with 3 years of data collapsed into 1 row using weighting
 - Weighted player stats by recency (5, 4, 3 weighting)
 - This way, more recent performance is taken into greater account, similar to how GMs might make real contract decisions

Data Processing

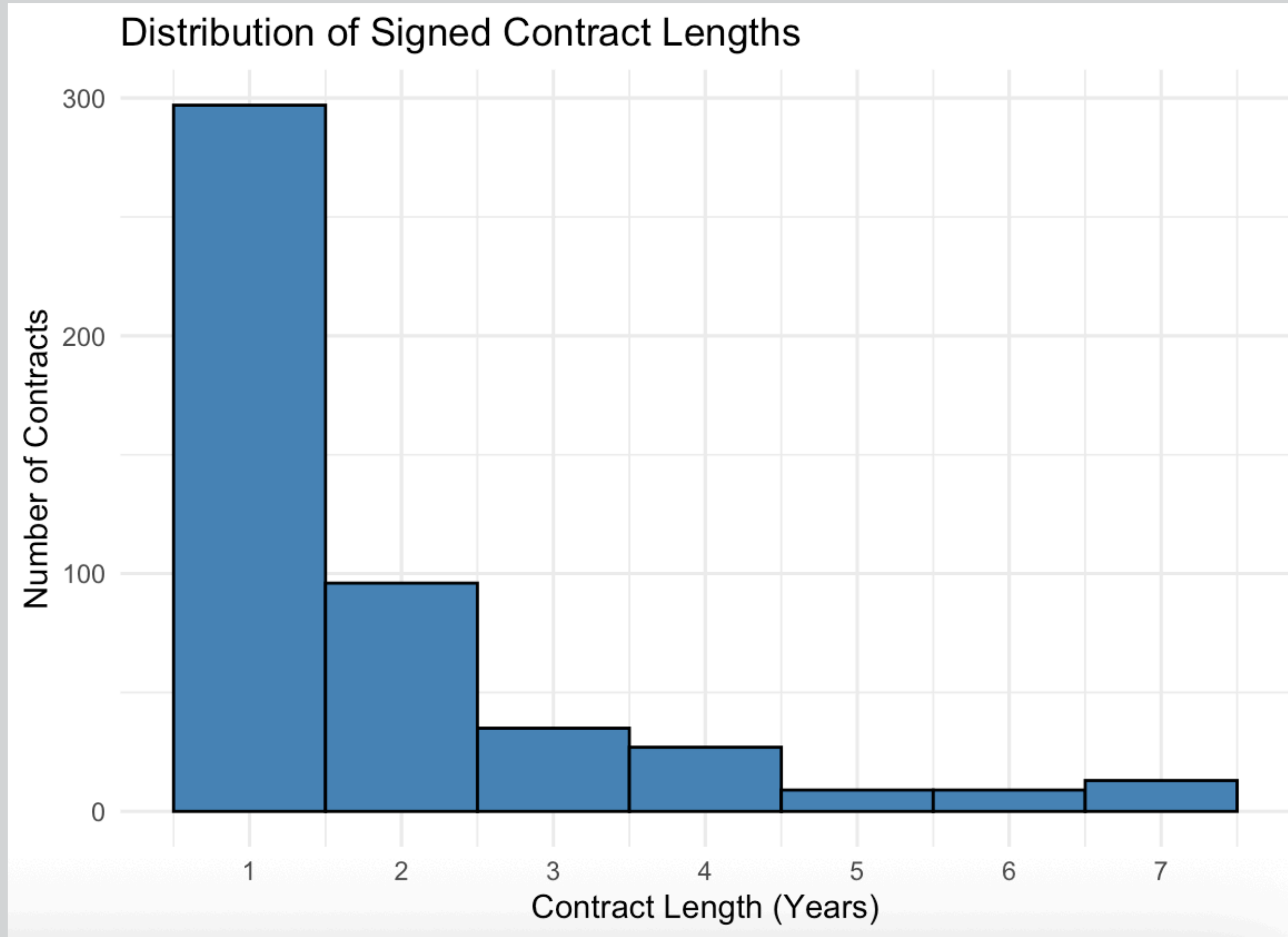


	playerId	name	Age	Pos	season	GP	G	A	PTS	plus_minus	PIM	gameScore	I_F_xGoals
1	8476878	Zemgus Girgensons	28	C	2021	56	10	8	18	-7	27	18.99	8.89
2	8476878	Zemgus Girgensons	29	C	2022	80	10	8	18	-5	14	25.81	13.55
3	8476878	Zemgus Girgensons	30	C	2023	63	8	6	14	-1	21	16.78	8.12

	playerId	name	Age	Position	GP_wgt	G_wgt	A_wgt	PTS_wgt	plus_minus_wgt	PIM_wgt	gameScore
1	8476878	Zemgus Girgensons	30	C	66.91667	9.166667	7.166667	16.333333	-3.833333	20.16667	20.3425



Signed Contract Imbalance



- 80% of NHL contracts in my data set are 1-2 years
- Early models severely under-predicted 3-7 year contracts
- Tried fixes like weighted models, SMOTE, and random oversampling

Signed Contract Imbalance

Train Data Distribution

	Signed_Length	n	Percentage
	<dbl>	<int>	<dbl>
1	1	297	61.1
2	2	96	19.8
3	3	35	7.2
4	4	27	5.6
5	5	9	1.9
6	6	9	1.9
7	7	13	2.7

2025 prediction distribution

	Predicted_Contract_Term	n	Percentage
	<chr>	<int>	<dbl>
1	Long	12	7.3
2	Medium	11	6.7
3	Short	142	86.1

Brief Overview of Models



Contract Term Model – Ranger Multiclass classification model

- Goal: predict the players' predicted contract length
 - Originally attempted to predict actual years, pivoted to a binned term approach due to early models poorly predicting 3 - 7 year contracts
 - “Short” = 1 - 2 year contract length
 - “Medium” = 3 - 4 year contract length
 - “Long” = 5+ year contract length
 - Model predicts Predicted_Contract_Term to be fed into the AAV model as a feature

Brief Overview of Models



Variables Used in Contract Term Model – (29 total variables)

- **Standardized Statistics (per 60 mins)**
 - EVG, PPG, TAKE, GIVE, BLK, HIT, Primary Assists
- **Traditional Statistics**
 - GP, +/-, ATOI, SPCT, FO%
- **Advanced Metrics**
 - gameScore, iceTimeRank, FlurryAdjustedxGoals/60, FlurryAdjustedxGoals Against/60
- **Others**
 - Previous Cap% and Length, Captaincy, Draft Round, Age 32+ dummie, Interaction Terms

Term Model Results



Model Evaluation

- Accuracy – .8487
 - Not significantly better than “no-information” rate **but** has predictive qualities
- Kappa – .3964
 - Indicates fair/moderate agreement beyond chance
- Balanced Accuracy
 - Indicates Long Class is the strongest
 - Rare cases are often accurate
 - Short Class is over predicted (.6701)
 - Struggles to predict Medium Term Contracts

Statistics by Class:			
	Class: Long	Class: Medium	Class: Short
Sensitivity	0.57143	0.21429	0.9592
Specificity	0.98214	0.97143	0.3810
Pos Pred Value	0.66667	0.50000	0.8785
Neg Pred Value	0.97345	0.90265	0.6667
Prevalence	0.05882	0.11765	0.8235
Detection Rate	0.03361	0.02521	0.7899
Detection Prevalence	0.05042	0.05042	0.8992
Balanced Accuracy	0.77679	0.59286	0.6701

Term Model Takeaways



- **The contract term model is accurately predicting Long-Term contracts, but over-predicting Short-Term Contracts**
- **The model has a difficult time identifying medium-term contracts**
- **Some players are receiving unrealistic contracts, Age isn't being punished enough**
- **Top 5 Variables Ranked By Importance**
 - $ATOI \times GameScore$ (and their individual vars)
 - plus_minus
 - prev_cap_pct
 - PIMs
 - $FlurryAdjusted \times Goals/60$

Brief Overview of Models



AAV Model – Cubist Model (Regression)

- Goal: Predict a players' percentage of the cap with predicted term taken into account
 - Attempted Linear regression, Random Forest, Ranger, Cubist
 - Non-numeric variables had to be swapped for dummie variables
 - ex. FA_typeRFA and FA_typeUFA

AAV Model Output is then combined with the Term Model results to give the full predicted player contracts

Brief Overview of Models



Variables Used in AAV Model – (35 total variables)

- **Standardized Statistics (per 60 mins)**
 - EVG, PPG, EV, PP, HIT, PenaltiesDrawn
- **Traditional Statistics**
 - ATOI, PIMS, FO%, SPCT
- **Advanced Metrics**
 - On-ice Goals for/60, FlurryAdjustedxGoalsAgainst/60
- **Others**
 - Predicted Contract Term, Previous Contract Details, Captain, Age 32+ (Draft Round Found Insignificant)

AAV Model Results



Model Evaluation

- R-squared: 0.73716
 - 74% of variance in AAV is captured by my model
- RMSE: 0.01198
 - On average predictions can be off by ~\$1,100,000
- MAE: 0.0049
 - Average error is ~ \$470,000, Indicates that most are close with few outliers

AAV Model Takeaways



AAV predictions are relatively tight, with outliers skewing prediction accuracy

- The average error is Less than \$500,000

AAV predictions aren't punishing age enough – older players are being predicted larger contracts than I would expect

Top 5 Variables Ranked By Importance

- ATOI x GameScore (and their individual vars)
- plus_minus
- prev_cap_pct
- Contract_Term_short dummie – Indicates AAV decrease if contract is short
- PP/60

Final Results: Mitch Marner



Age

- 26 years old

Previous Contract

- 6 years, \$10.9m AAV (13.4% cap)

2025 Statistics

- 27 G, 77 A, 102 Pts
- 33 PP pts
- +/- : 18
- ATOI: 21:18

Predicted Contract

- Long-Term (5+)
- 14.9% Cap
- \$14,271,059/yr



Final Results: Evan Bouchard

Age

- 25 years old

Previous Contract

- 2 years, \$3.9m AAV (4.7% cap)

2025 Statistics

- 14 G, 53 A, 67 Pts
- 26 PP pts
- +/- : 14
- ATOI: 23:27

Predicted Contract

- Long-Term (5+)
- 10.6% Cap
- \$10,159,977/yr



15 Most Expensive FA Contracts



name	Age	prev_Length	prev_Cap_Pct	Predicted_Contract_Term	Predicted_Cap_Pct	Predicted_AAV
mitsch marner	27	6	0.133779141	Long	0.149435177	\$14,271,059
aaron ekblad	28	8	0.100000000	Long	0.117325515	\$11,204,587
evan bouchard	25	2	0.046706587	Long	0.106387198	\$10,159,977
brent burns	39	8	0.106666667	Short	0.101121075	\$9,657,063
john tavares	34	7	0.138364780	Medium	0.095969342	\$9,165,072
dmitry orlov	33	2	0.092814371	Long	0.094646081	\$9,038,701
brock nelson	33	6	0.073619632	Long	0.087173946	\$8,325,112
brad marchand	36	8	0.081666667	Short	0.086072639	\$8,219,937
nikolaj ehlers	28	7	0.075471698	Long	0.084425971	\$8,062,680
sam bennett	28	4	0.054294479	Medium	0.079902925	\$7,630,729
claudio giroux	37	3	0.078787879	Short	0.074127406	\$7,079,167
brock boeser	27	3	0.080606061	Long	0.070188835	\$6,703,034
vladislav gavrikov	29	2	0.070359281	Long	0.068428949	\$6,534,965
juraj slafkovsky	20	3	0.053939394	Short	0.065008886	\$6,208,349
kyle palmieri	33	4	0.061349693	Medium	0.063816763	\$6,094,501

Highest Increases in Cap%



	▲	name	Age	prev_Cap_Pct	Predicted_Cap_Pct	AAV_change
1		evan bouchard	25	0.04671	0.10639	0.05968
2		jj peterka	23	0.01084	0.05743	0.04660
3		matthew knies	22	0.01121	0.05210	0.04089
4		luke hughes	21	0.02242	0.06031	0.03789
5		tyson foerster	23	0.01046	0.04198	0.03152
6		marco rossi	23	0.02077	0.05219	0.03142
7		luke evangelista	22	0.01000	0.04118	0.03118
8		morgan geekie	26	0.02395	0.05193	0.02798
9		sam bennett	28	0.05429	0.07990	0.02561
10		jonathan drouin	29	0.02841	0.05345	0.02504

Highest Decreases in Cap%



	▲	name	Age	prev_Cap_Pct	Predicted_Cap_Pct	AAV_change
1		jamie benn	35	0.12667	0.06198	-0.06468
2		john tavares	34	0.13836	0.09597	-0.04240
3		jeff petry	37	0.07669	0.03537	-0.04132
4		christian dvorak	28	0.05460	0.01586	-0.03875
5		ivan provorov	28	0.08282	0.05045	-0.03237
6		robby fabbri	29	0.04848	0.01702	-0.03146
7		david savard	34	0.04294	0.01220	-0.03075
8		patrick kane	36	0.07386	0.04434	-0.02952
9		alec martinez	37	0.04545	0.01633	-0.02913
10		joel armia	31	0.04172	0.01573	-0.02599

Model Limitation/ Improvements



- Models are built on only 5 years' worth of Free Agent data – Introducing more would be helpful for prediction
- Predicted Contract Term rather than an exact Length
- Model is over-predicting “Short” contracts, and under-predicting Medium Contracts
 - Using more data, including more intangible factors and interactions, could improve this problem
- Model doesn't value defensive contribution as much as it does offensive production
 - Some Defensive player contracts are lower than what I expected

Model Limitation/ Improvements



- Models don't take into account the team strength or cap spent, which can be big factors in contract negotiation
- Models don't take intangible player factors into account
 - tested Captain status, Draft Round, and FA type – Some significance in models
 - Would include next time: Award/Cup winner Vars, “Hometown Team”, Loyalty, player roles, playoff data
- Models are not punishing Old Age enough in AAV and Term prediction
 - Would add more interaction terms with Age

Thank You!

Questions?

